A Cross-Polytope Geometry of the Primes: Orbit Minima, Exact Residue Counts, and Singular Series

The 16-Cell at Modulus 30, Four Structural Theorems, and the Local Factors of the Classical Prime Conjectures

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Abstract

Removing the multiples of 2, 3 and 5 leaves eight residue classes modulo 30. Pairing every class a with its mirror $30 - a$ turns these eight classes into the eight vertices of the 16-cell, the four-dimensional cross-polytope, and the same construction at every primorial $M_k = 2{\cdot}3 \cdots p_k$ produces a cascade of cross-polytopes $\beta_{\varphi(M_k)/2}$. On this cascade we study the action of the sign-flip group $G_k \cong (\mathbb{Z}/2)^{k-1}$ and establish four structural results: the orbits are exactly the square classes of the unit group; for $k \geq 6$ the orbit minima up to p_{k+1}^2 are precisely the primes in (p_k, p_{k+1}^2) together with p_{k+1}^2, and this threshold is sharp; the cascade is recursive, with p_{k+1} the smallest non-trivial orbit minimum; and the number of orbits grows by the exact factor $(p_{k+1}-1)/2$ per level. We then apply one common scheme to twin primes, Polignac, Goldbach, Sophie Germain primes, Hardy–Littlewood constellations, primes of the form n^2+1, arithmetic progressions, exponential families and the Collatz iteration. For each problem the geometry gives an exact Chinese-remainder count of admissible configurations, a truncated singular series that converges to the classical Hardy–Littlewood constant, a conditional statement, the strongest known unconditional result, and numerical checks up to 10^8. The construction determines all local factors; whether every admissible configuration is actually occupied by primes (the *lifting problem*) is not decided by it and remains open wherever it is open classically.

Keywords: cross-polytope; 16-cell; primorial; unit group; sign-flip action; orbit minima; singular series; Hardy–Littlewood; Goldbach; prime k-tuples.
MSC 2020: 11N05 (primary); 11N13, 11P32, 11N35, 52B11 (secondary).

How to read this paper. *Narrative path:* every section opens with a shaded box "In plain words" that explains the idea without formulas; together with the figures these boxes form a readable summary of the whole paper. *Technical path:* definitions, theorems and proofs, with full proofs either in place or in Appendix A. **Status marks: [S]** proved in full (here or in the cited literature); **[G]** exact geometric reformulation that makes no existence claim; **[N]** numerical finding with the stated range; **[H]** heuristic; **[O]** open.

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Introduction

In plain words

Every prime larger than 5 leaves one of only eight possible remainders when divided by 30. These eight remainders come in four mirror pairs (1 and 29, 7 and 23, ...), and four mirror pairs are exactly what one needs to build the simplest "four-dimensional diamond", the 16-cell. This paper takes that observation seriously: it shows what the diamond and its larger relatives at 210, 2310, ... can *prove* about primes, and where their power ends. The short answer is: the geometry computes all the "local" information exactly, i.e. how often a pattern of primes is *allowed* by small divisors; it cannot by itself show that the allowed patterns are actually *realised* infinitely often. That last step is the core of the famous open problems, and we say so in every case.

From a game to a polytope. The series of papers summarised here began with a simple bookkeeping game in which matchsticks are laid on the integers $1, 2, 3, \ldots$ [31]. In this game the primes are exactly the moves that open a new direction. Sieving out the first three such directions $(2, 3, 5)$ leaves eight residue classes modulo 30, and the natural pairing $a \leftrightarrow 30 - a$ makes them the vertex set of the 16-cell [35]. The same recipe at every primorial gives a cascade of cross-polytopes [36, 47]; a German book-length account is [50]. Figure 1 shows the path.

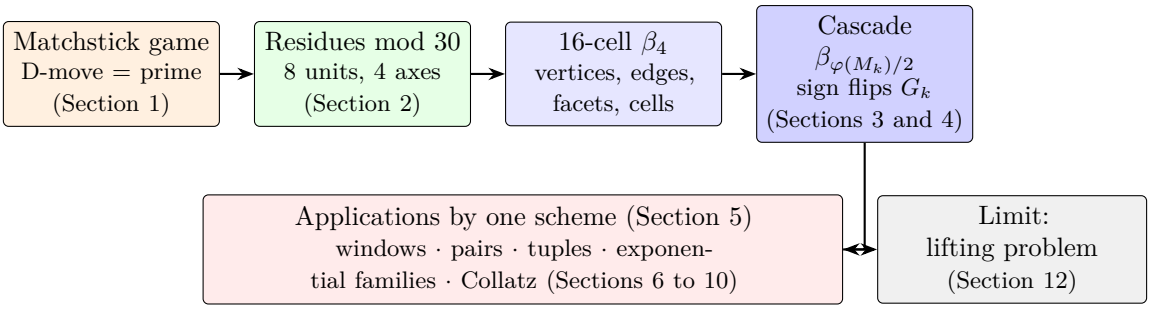


Figure 1: Overview. The game singles out the primes; three sieve steps leave the eight units modulo 30; the mirror pairing makes them the 16-cell; the same construction at every primorial gives a cascade on which four structural theorems hold; one application scheme then treats the classical problems and isolates the single step it cannot perform.

What is proved and what is not. All statements marked **[S]** are theorems with complete proofs. Most of them are elementary (Chinese remainder theorem, structure of unit groups, Mertens' theorem); their value lies in the uniform geometric organisation and in a few sharp statements that we have not found in the literature in this form, most notably the *completeness theorem* (Theorem 4.5) with its sharp threshold $k \geq 6$ (Proposition 4.6). For every classical problem we separate four layers: the exact local count (**[S]**), the classical conditional asymptotic (Hardy–Littlewood type), the strongest unconditional result from the literature, and our own numerics (**[N]**). The geometric constructions do not replace the analytic input that the open problems require, and we make no claim that they do.

Contributions.

1. A self-contained account of the 16-cell model of the units modulo 30, including its complete incidence spectrum (Section 2).
2. A uniform treatment of the sign-flip action on $U(M_k)$: orbits are square classes, the group is the 2-torsion of the unit group, the Cayley graph is a disjoint union of hypercubes, and the averaging operator has an explicit binomial spectrum (Sections 3 and 4).
3. A short rigorous proof of the completeness theorem via one divisibility lemma ($4M_k \mid n^2 - z^2$), with the sharp threshold and the identification of p_{k+1}^2 as the smallest composite orbit minimum (Section 4).
4. A six-step application scheme and its execution for the classical prime problems, with explicit residue sets, exact counts, constants and numerics (Sections 5 to 9).
5. A clear statement of the limit of the method (non-locality and the lifting problem), supported by an explicit count of "local" versus true orbit minima (Section 12).

Notation. $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$ are the primes in increasing order, $M_k = p_1 p_2 \cdots p_k$ is the k-th primorial ($M_3 = 30$, $M_4 = 210$, $M_5 = 2310$), $U(M) = (\mathbb{Z}/M\mathbb{Z})^\times$, φ is Euler's function, and units modulo M are represented by integers in $[1, M]$. β_d denotes the d-dimensional cross-polytope (convex hull of $\pm e_1, \ldots, \pm e_d$). A glossary for non-specialists is in Appendix D.

1 The matchstick game and the notion of a prime

In plain words

Imagine walking along the numbers $2, 3, 4, \ldots$ and laying down one matchstick at each number. A stick is of one of three kinds. If the number can be written as a product of two *different* smaller numbers ($12 = 3 \cdot 4$), the stick "closes a rectangle". If it is the square of a prime ($25 = 5 \cdot 5$), it is an "echo". Otherwise nothing old explains the number, and it opens a *new direction*. The new directions are exactly the primes. Nothing deep happens here; the game is a bookkeeping device. Its use is that it turns multiplication into geometry: each prime is an axis, each number a point in a lattice with one axis per prime, and many classical facts (Euler's product, Mertens' constant, Ramanujan's highly composite numbers) become statements about this lattice.

1.1 Rules

We use the following binding formulation of the rules of [31]. For $n \geq 2$ the move at n is

- **S** (*closing*) if $n = ab$ with integers $1 < a < b$;
- otherwise **E** (*echo*) if $n = p^2$ for a prime p;
- otherwise **D** (*new direction*).

The priority $S > E > D$ is part of the rule. Figure 2 shows the board for $2 \leq n \leq 40$.

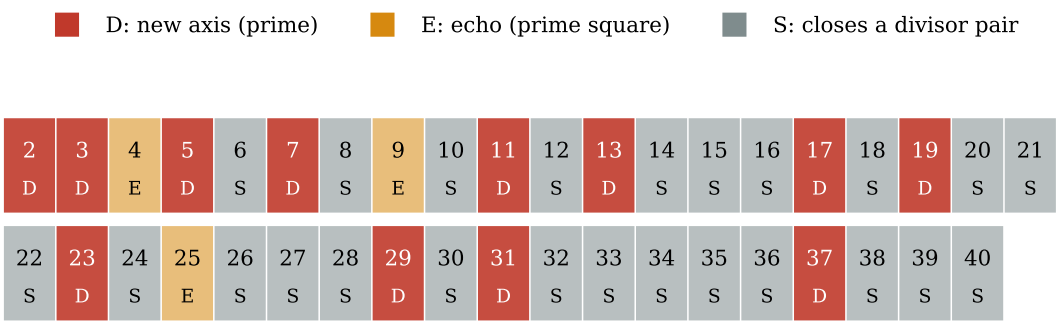


Figure 2: The matchstick board for $2 \leq n \leq 40$. Red: D-moves (new direction), yellow: E-moves (prime squares $4, 9, 25$), grey: S-moves (closing a divisor pair). The red cells are exactly the primes (Proposition 1.1).

1.2 Game primes are primes

Proposition 1.1 ([S]). *For $n \geq 2$ the move at n is D if and only if n is prime.*

Proof. A prime has no factorisation $n = ab$ with $1 < a \leq b$, so neither S nor E applies. Conversely let n be composite with smallest prime factor q, and write $n = qm$ with $m \geq q$. If $m > q$, the move is S; if $m = q$, then $n = q^2$ and the move is E. In neither case is it D. □

1.3 What the game makes visible

In plain words: the lattice of exponents

If every prime is an axis, then every number is a point whose coordinates are its prime exponents: $12 = 2^2 \cdot 3$ sits at height 2 on axis 2 and height 1 on axis 3. The total number of sticks is the number of prime factors counted with multiplicity.

Write $n = \prod_p p^{a_p}$. The map $n \mapsto (a_p)_p$ identifies the positive integers with the finitely supported points of $\mathbb{N}_0^{\{\text{primes}\}}$; $\Omega(n) = \sum_p a_p$ and $\omega(n) = \#\{p : a_p > 0\}$ count all sticks and the occupied axes. Three classical facts take a geometric form **[G]**:

1. *Euler's product.* Summing n^{-s} over the lattice factorises over the axes: $\sum_{n\geq 1} n^{-s} = \prod_p (1 - p^{-s})^{-1}$ for $\Re s > 1$.
2. *Mertens.* The proportion of integers with no stick on the axes $p \leq x$ is $\prod_{p\leq x}(1 - 1/p) \sim e^{-\gamma}/\log x$ [18]; this is the density that reappears in every constant of Sections 7 and 8.
3. *Highly composite numbers.* A number n is highly composite if it has more divisors than every smaller number [24]. In the lattice these are the points that spread their sticks most efficiently over the lowest axes; their exponents are non-increasing in p. Up to 10^4 there are 20 of them (Figure 3); their distribution was determined by Nicolas [20], see also [33].

The primorials M_k themselves are the "one stick per axis" points. A famous criterion of Nicolas [21] states that the Riemann hypothesis holds if and only if $M_k/\varphi(M_k) > e^\gamma \log\log M_k$ for all $k \geq 1$. We do not use this criterion, but it shows that the primorial skeleton on which the rest of the paper is built carries deep information.

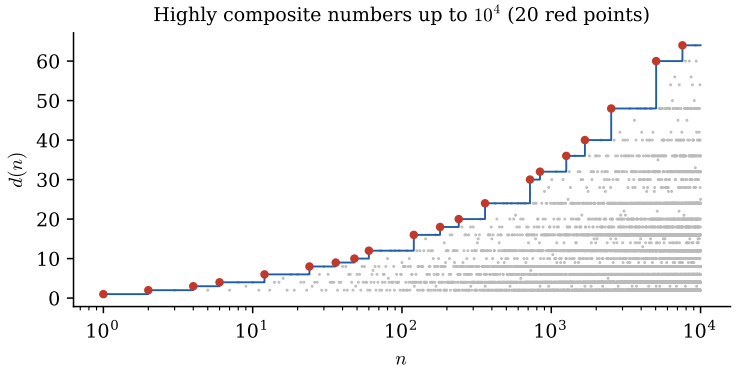


Figure 3: Divisor function $d(n)$ (grey) and the staircase of record values; the 20 highly composite numbers up to 10^4 are marked red.

1.4 The adversarial version

The *Devil's game* (the sum-and-product puzzle of Freudenthal) asks for pairs (x, y) that two players can identify from the sum and the product only after a prescribed exchange of statements. In [38] this puzzle is read inside the game: every statement of the puzzle is a statement about which factorisations of the product (S-moves) and which decompositions of the sum remain possible, and the solutions are governed by D-moves (primes) and pure powers of 2. Exhaustive search finds 18 solutions with components up to 99 999 **[N]** [38]. We mention the puzzle only as an illustration that "being a D-move" is a robust, game-theoretically usable notion; nothing later depends on it.

2 Residues modulo 30 and the 16-cell

In plain words

Throw away every number divisible by 2, 3 or 5. What is left, modulo 30, are eight remainders: $1, 7, 11, 13, 17, 19, 23, 29$. Every prime above 5 has one of them. These eight come in four mirror pairs that add up to 30. Put each pair on one axis of four-dimensional space, one remainder at $+1$ and its mirror at -1, and you get eight points: the corners of the 16-cell, the four-dimensional analogue of

the octahedron. Two corners are joined by an edge unless they are mirror images. Edges, triangles and tetrahedra of this object then carry arithmetic meaning: for example, the number of edges whose two corners add up to a given remainder is exactly the local factor that appears in the Goldbach conjecture.

2.1 Three sieve steps

Proposition 2.1 (**[S]**). *$U(30) = \{1, 7, 11, 13, 17, 19, 23, 29\}$, $\varphi(30) = 8$, and every prime $p > 5$ lies in one of these classes. Moreover $U(30) \cong \mathbb{Z}/2 \times \mathbb{Z}/4$: the elements $7, 13, 17, 23$ have order 4, and $11, 19, 29$ have order 2.*

Proof. The first statement is the sieve by $2, 3, 5$. By the Chinese remainder theorem $U(30) \cong U(2) \times U(3) \times U(5) \cong \mathbb{Z}/2 \times \mathbb{Z}/4$. Direct computation: $7^2 = 49 \equiv 19$, $13^2 \equiv 19$, $17^2 \equiv 19$, $23^2 \equiv 19$, while $11^2 \equiv 19^2 \equiv 29^2 \equiv 1 \pmod{30}$. □

2.2 Mirror pairing and axes

The map $a \mapsto 30 - a$ ($a \mapsto -a$) is a fixed-point-free involution on $U(30)$. Its orbits are the four *axes*

$$A_1 = \{1, 29\}, \quad A_2 = \{7, 23\}, \quad A_3 = \{11, 19\}, \quad A_4 = \{13, 17\}.$$

Definition 2.2. Identify the smaller element of A_i with $+e_i$ and the larger with $-e_i$ in $\mathbb{R}^4$. The convex hull of these eight points is the 16-cell β_4. We write A_i^+ and A_i^- for the two half-axes.

In β_d two vertices span an edge if and only if they are not antipodal; $k+1$ vertices span a k-face if and only if no two are antipodal. Hence β_4 has f-vector $(8, 24, 32, 16)$: 8 vertices, $\binom{8}{2} - 4 = 24$ edges, $\binom{4}{3}2^3 = 32$ triangular facets and $2^4 = 16$ tetrahedral cells, each cell choosing one vertex from every axis.

2.3 Incidence spectrum

Proposition 2.3 (**[S]**). *For the sums of vertex labels modulo 30:*

1. *every axis sums to 0;*
2. *the 24 edge sums are never 0; the classes $6, 12, 18, 24$ occur three times each, $10, 20$ twice, and the remaining eight even classes once;*
3. *the 32 facet sums are never 15; the classes 5 and 25 occur four times each, every other odd class twice;*
4. *the 16 cell sums, reduced modulo 30 and represented in $(0, 30]$, take each even value $2, 4, \ldots, 28$ once and the value 30 twice.*

Proof. A finite check of $4 + 24 + 32 + 16$ cases, reproduced in Appendix B; the distribution is shown in Figure 4. □

The edge spectrum is not a curiosity. For an even residue r let $g_{30}(r)$ be the number of *ordered* pairs $(a, b) \in U(30)^2$ with $a + b \equiv r \pmod{30}$.

Theorem 2.4 (Local Goldbach factor at 30, **[S]**).

$$g_{30}(r) = \begin{cases} 8, & r \equiv 0 \pmod{30}, \\ 6, & 6 \mid r,\ 30 \nmid r, \\ 4, & 10 \mid r,\ 30 \nmid r, \\ 3, & \textit{otherwise}, \end{cases} \qquad \textit{equivalently} \qquad g_{30}(r) = \prod_{p \in \{3,5\}} N_p(r),$$

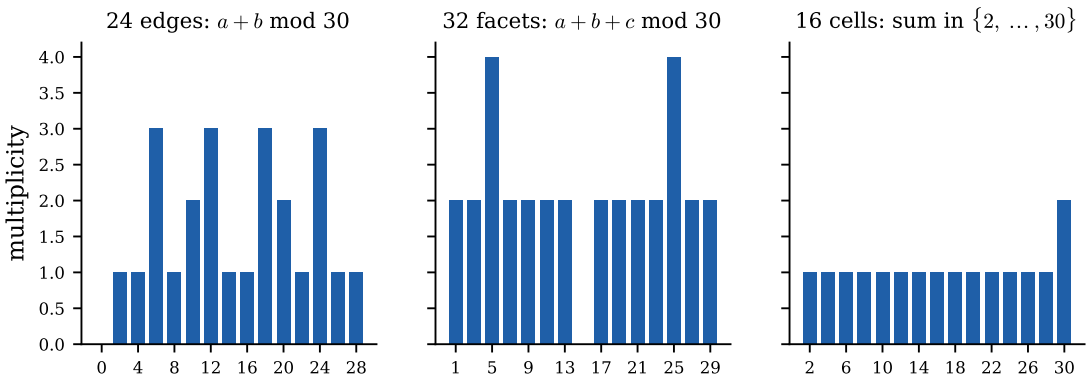


Figure 4: Sum distributions of the 24 edges, 32 facets and 16 cells of the 16-cell modulo 30 (Proposition 2.3). Edge sums avoid 0 (that is the axis value), facet sums avoid 15.

with $N_p(r) = p-1$ *if* $p \mid r$ *and* $N_p(r) = p-2$ *otherwise. Geometrically:* $g_{30}(0) = 2 \cdot \#\{\text{axes}\}$*, and for* $r \not\equiv 0$*,* $g_{30}(r) = 2 \cdot \#\{\text{edges with sum } r\} + \#\{a : 2a \equiv r\}$*, the last term being the "diagonal" pairs* (a, a)*.*

Proof. By the Chinese remainder theorem the number of pairs factorises over $p = 2, 3, 5$. For $p = 2$ both summands are odd, giving one pair modulo 2 for even r. For odd p, the pairs (a, b) of non-zero residues with $a + b \equiv r$ number $p - 1$ if $r \equiv 0$ (every $a \neq 0$ works) and $p - 2$ otherwise ($a \neq 0$ and $a \neq r$). The geometric decomposition is immediate: if $r \equiv 0$ the pairs are $(a, -a)$, i.e. the four axes in both orders; if $r \not\equiv 0$ an unordered pair $\{a, b\}$ with $a \neq b$ is non-antipodal, hence an edge, and contributes two ordered pairs. Comparing with Proposition 2.3 gives the table (e.g. for $r = 2$ there is the single edge $\{13, 19\}$ and the diagonal pair $(1, 1)$, so $g_{30}(2) = 2 \cdot 1 + 1 = 3$). □

Remark 2.5 (a negative result). The analogous attempt for sums of three units fails: the ordered triples with sum in the odd classes $1, 5, 3, 15$ number $39, 36, 26, 24$, which is *not* proportional to the facet multiplicities of Proposition 2.3. The local factor of the ternary Goldbach problem is still a product of local counts (Section 7), but it is not a face count of β_4.

2.4 Two great squares

Proposition 2.6 (**[S]**). *The solutions of* $x^2 \equiv 1 \pmod{30}$ *form the subgroup* $T_1 = U(30)[2] = \{1, 11, 19, 29\} \cong (\mathbb{Z}/2)^2$*, and* $T_2 = 7T_1 = \{7, 13, 17, 23\}$ *is its other coset. Geometrically* $T_1 = A_1 \cup A_3$ *and* $T_2 = A_2 \cup A_4$ *are two orthogonal great squares of* β_4 *(each a* β_2 *lying in a coordinate plane).*

Proof. The squares were computed above. T_1 consists of the two axes A_1, A_3, so its convex hull is the square with vertices $\pm e_1, \pm e_3$; likewise for T_2. □

Remark 2.7. In earlier papers of the series [35, 36] T_1, T_2 were called a "stella octangula" decomposition. Since each T_i contains antipodal pairs, the sets are great squares and not cells; the decomposition is the splitting of $\mathbb{R}^4$ into two orthogonal coordinate planes.

2.5 Why the 16-cell

There are exactly six regular convex polytopes in dimension four: the 5-cell, the tesseract, the 16-cell, the 24-cell, the 120-cell and the 600-cell [5]. The choice of the 16-cell is a modelling choice, which we state openly. It rests on three requirements: (i) eight vertices, one per unit; (ii) a fixed-point-free central symmetry that realises $a \mapsto -a$; (iii) the edge relation should be

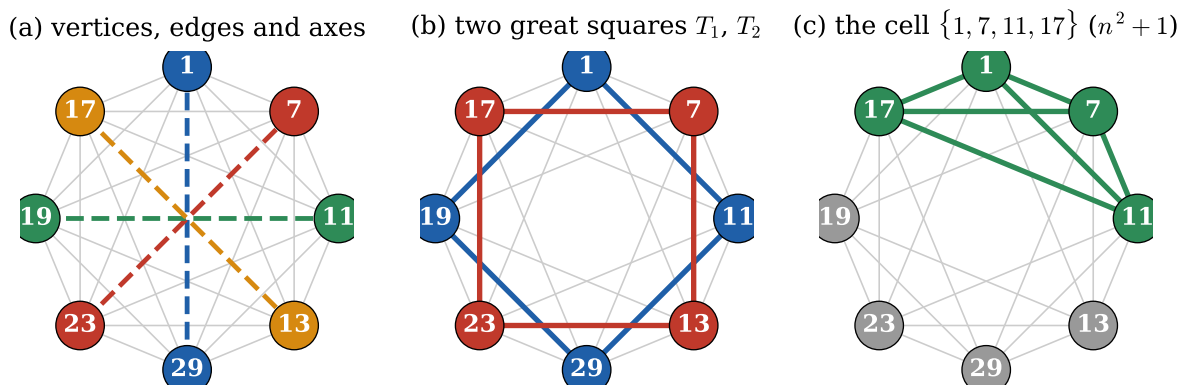


Figure 5: The 16-cell in the standard octagon projection. Left: the four axes $A_1, \ldots, A_4$ (dashed) and all 24 edges. Middle: the great squares $T_1 = \{1, 11, 19, 29\}$ (the orbit of 1 under sign flips) and $T_2 = \{7, 13, 17, 23\}$. Right: the cell $\{1, 7, 11, 17\}$, which carries all primes of the form $n^2 + 1$ beyond 5 (Theorem 8.2).

determined by this symmetry alone. Among the regular four-dimensional polytopes only the 16-cell has eight vertices, and in every dimension the cross-polytope is the unique polytope whose graph is "all pairs except antipodes" (the cocktail-party graph). For general M_k the same requirements lead to $\beta_{\varphi(M_k)/2}$ (Section 3). Exploratory checks at 210 and 2310 gave no indication of additional exceptional structure (for instance root systems of type F_4 or E_8) **[N]**; we use none.

2.6 Fermat purity of 30

Proposition 2.8 (**[S]**). *$U(M_k)$ is a 2-group if and only if $k \le 3$. Equivalently, the cross-polytope dimension $\varphi(M_k)/2$ is a power of two exactly for $k \le 3$, and $\varphi(M_k)/2 = k + 1$ holds only for $k = 3$.*

Proof. $|U(M_k)| = \prod_{j \le k}(p_j - 1)$ is a power of 2 iff every $p_j \le p_k$ is of the form $2^m + 1$; $2, 3, 5$ are, 7 is not. The values $\varphi(M_k)/2$ are $1, 4, 24, 240, \ldots$ for $k = 2, 3, 4, 5, \ldots$ and grow faster than $k + 1$. □

So 30 is the last primorial whose unit group is built only from the Fermat primes $3, 5$; this is why the 16-cell carries the clean $\mathbb{Z}/2 \times \mathbb{Z}/4$ structure without odd-order components.

2.7 The address of a prime

Definition 2.9. For a prime $p > 5$ let k be the smallest index with $p < M_k$. The *address* of p is $(k, A_i, \pm, s)$, where $A_i^{\pm}$ is the half-axis containing p mod 30 and s is the 0-based position of p among the units in $[1, M_k]$ that lie on the same half-axis (there are $\varphi(M_k)/8$ such slots).

Examples: $7 \mapsto (3, A_2, +, 0)$, $31 \mapsto (4, A_1, +, 1)$, $101 \mapsto (4, A_3, +, 3)$. Further addresses are listed in Appendix C.

Proposition 2.10 (**[S]**). *Distinct primes $p > 5$ have distinct addresses.*

Proof. The address determines k, the residue p mod 30 and the position among the units below M_k in that class, hence the integer $p < M_k$ itself. □

A second, classical label refines the half-axis **[S]**: a prime $p \equiv 1 \pmod 4$ is a sum of two squares (Fermat), $p \equiv 3 \pmod 8$ is a sum of three but not two squares, and $p \equiv 7 \pmod 8$ needs four squares (Legendre, Lagrange) [11]; see [34] for the square atlas. We call these the square classes A, B, C.

3 From 30 to every primorial: the cross-polytope cascade

In plain words

Nothing about the construction is special to 30. Remove the multiples of $2, 3, 5, 7$ and you are left with 48 remainders modulo 210; pair them with their mirrors and you get a 24-dimensional cross-polytope. Each new prime multiplies the number of corners by $p - 1$. On each of these polytopes there is a natural set of "switches": for every odd prime $p_j \le p_k$ one may flip the sign of a remainder with respect to p_j alone, leaving it unchanged modulo all the other primes. Flipping all switches at once gives the mirror image. A group of remainders that can be reached from one another by flipping switches is called an *orbit*, and its smallest member is its *orbit minimum*. Orbit minima are the central objects of the paper; the next section shows that, in a precisely described range, they are exactly the primes.

3.1 Setting

Fix $k \ge 2$. The vertex set of level k is $U(M_k)$, of size $\varphi(M_k) = \prod_{j \le k}(p_j - 1)$. Because M_k is even, $a \mapsto M_k - a$ is a fixed-point-free involution on $U(M_k)$ ($a = M_k/2$ is not a unit). Its $\varphi(M_k)/2$ orbits are the *axes* of level k, and as in Definition 2.2 the units become the vertices of $\beta_{\varphi(M_k)/2}$.

3.2 Sign-flip action

For $2 \le j \le k$ let $e_j \in U(M_k)$ be the unique unit with $e_j \equiv -1 \pmod{p_j}$ and $e_j \equiv 1 \pmod{p_i}$ for $i \ne j$, and define $\sigma_j(a) = e_j a \bmod M_k$. Thus σ_j flips the sign of a modulo p_j and changes nothing else. Let $G_k = \langle \sigma_2, \dots, \sigma_k \rangle$.

Proposition 3.1 ([S]). *1. $G_k \cong (\mathbb{Z}/2)^{k-1}$, and G_k acts on $U(M_k)$ by multiplication with the subgroup $U(M_k)[2] = \{x : x^2 \equiv 1\} = \{\prod_{j \in S} e_j : S \subseteq \{2, \dots, k\}\}$.*

2. The action is free.

3. The product of all σ_j is the antipodal map $a \mapsto -a$.

Proof. By the Chinese remainder theorem $U(M_k) \cong \prod_{j \le k} U(p_j)$, and for odd p_j the cyclic group $U(p_j)$ has 2-torsion $\{\pm 1\}$, while $U(2)$ is trivial. Hence $U(M_k)[2] \cong (\mathbb{Z}/2)^{k-1}$ with basis $e_2, \dots, e_k$. A multiplication by a group element $g \ne 1$ has no fixed point, so the action is free. Finally $\prod_j e_j \equiv -1$ modulo every odd p_j and modulo 2. □

3.3 Orbit criterion

Proposition 3.2 ([S]). *Two units y, z lie in the same G_k-orbit if and only if $y^2 \equiv z^2 \pmod{M_k}$, equivalently $z \equiv \pm y \pmod{p_j}$ for every odd $p_j \le p_k$. Every orbit is a union of 2^{k-2} axes.*

Proof. $z \in yU(M_k)[2]$ iff $(z/y)^2 \equiv 1$ iff $z^2 \equiv y^2$. Modulo an odd prime, $z^2 \equiv y^2$ means $z \equiv \pm y$. By Proposition 3.1(3) every orbit is closed under $a \mapsto -a$; it has 2^{k-1} elements, hence 2^{k-2} axes. □

3.4 Counting

Definition 3.3. B_k is the set of orbit minima, i.e. the smallest representative in $[1, M_k]$ of every G_k-orbit.

Corollary 3.4 ([S]). *$|B_k| = \varphi(M_k)/2^{k-1}$, and every orbit minimum is $< M_k/2$. Explicitly $|B_k| = 2, 6, 30, 180, 1440, 12960$ for $k = 3, \dots, 8$.*

Proof. The action is free, so all orbits have 2^{k-1} elements. An orbit contains y and $M_k - y$, and one of them is $< M_k/2$. □

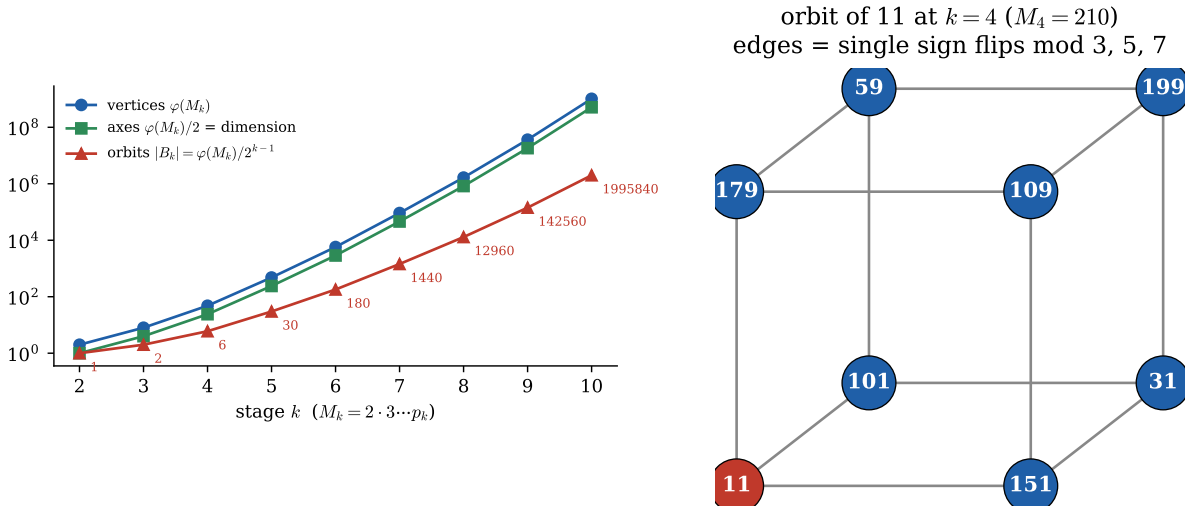


Figure 6: Left: growth of the cascade. Vertices $\varphi(M_k)$, cross-polytope dimension $\varphi(M_k)/2$ and number of orbits $|B_k|$ on a logarithmic scale. Right: the orbit of 11 modulo 210 drawn as the cube Q_3; each edge direction is one sign flip $\sigma_2, \sigma_3, \sigma_4$ (modulo $3, 5, 7$), and the orbit minimum 11 is marked red.

4 Four structural theorems

In plain words

Four facts organise the cascade. *Symmetry*: two remainders are in the same orbit exactly when their squares agree; the mirror is one of the switches. *Completeness*: from level $k = 6$ on (modulus 30030), every orbit minimum up to the square of the next prime is a prime, and every prime in that range is an orbit minimum, with exactly one exception, the square p_{k+1}^2 itself. Below level 6 this is false, and we show exactly where it fails. *Recursion*: the next prime is always the smallest orbit minimum larger than 1; so the machine "knows" the next prime. *Growth*: each new level multiplies the number of orbits by $(p-1)/2$. The heart of the proof of completeness is a single observation: two different numbers in the same orbit must have squares that differ by a multiple of $4M_k$, which is impossible if both are small.

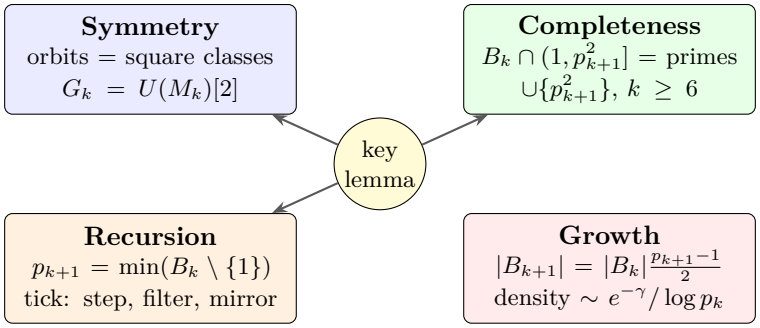


Figure 7: The four structural theorems. The key lemma (Lemma 4.2: $z \sim n \Rightarrow 4M_k \mid n^2 - z^2$) drives completeness and recursion; symmetry provides the orbit criterion on which everything rests.

4.1 Symmetry

Theorem 4.1 (Symmetry, **[S]**). *The G_k-orbits on $U(M_k)$ are the fibres of the squaring map $U(M_k) \to U(M_k)^2$; G_k is the 2-torsion subgroup $U(M_k)[2]$; the antipodal map is the all-flip element. In particular B_k is in bijection with the group of squares $U(M_k)^2$, and $|U(M_k)^2| = \varphi(M_k)/2^{k-1}$.*

Proof. This is Propositions 3.1 and 3.2 and Corollary 3.4 combined. □

4.2 Completeness

Lemma 4.2 (Key lemma, **[S]**). *Let $k \geq 2$ and let $z < n$ be units in the same G_k-orbit. Then $4M_k \mid n^2 - z^2$. Consequently every unit n with $n^2 \leq 4M_k$ is an orbit minimum.*

Proof. By Proposition 3.2 every odd $p_j \leq p_k$ divides $n^2 - z^2$, so $M_k/2 \mid n^2 - z^2$. Both n, z are odd, and the square of an odd number is $\equiv 1 \pmod 8$, so $8 \mid n^2 - z^2$. As $M_k/2$ is odd, $4M_k = 8 \cdot M_k/2$ divides $n^2 - z^2$. If $n^2 \leq 4M_k$ and some $z < n$ lay in the orbit of n, then $0 < n^2 - z^2 < n^2 \leq 4M_k$, a contradiction. □

Lemma 4.3 (**[S]**). *Every unit n modulo M_k with $1 < n \leq p_{k+1}^2$ is either a prime in (p_k, p_{k+1}^2) or equal to p_{k+1}^2.*

Proof. All prime factors of n are $\geq p_{k+1}$. If n has at least two of them (with multiplicity), then $n \geq p_{k+1}^2$, hence $n = p_{k+1}^2$. □

Lemma 4.4 (**[S]**). *$4M_k \geq p_{k+1}^4$ holds if and only if $k \geq 6$.*

Proof. For $k = 1, \ldots, 5$ the pairs $(4M_k, p_{k+1}^4)$ are $(8, 81)$, $(24, 625)$, $(120, 2401)$, $(840, 14641)$, $(9240, 28561)$. For $k = 6$: $4M_6 = 120120 \geq 17^4 = 83521$. If $4M_k \geq p_{k+1}^4$ for some $k \geq 6$, then by Bertrand's postulate $p_{k+2} < 2p_{k+1}$, so $p_{k+2}^4 < 16\, p_{k+1}^4 \leq p_{k+1} \cdot p_{k+1}^4 \leq p_{k+1} \cdot 4M_k = 4M_{k+1}$, using $p_{k+1} \geq 17 > 16$. □

Theorem 4.5 (Completeness, **[S]**). *For every $k \geq 6$,*

$$B_k \cap (1, p_{k+1}^2] \;=\; \{\, p \text{ prime} : p_k < p < p_{k+1}^2 \,\} \;\cup\; \{p_{k+1}^2\}.$$

In particular the orbit minima in the window $I_k = (p_k, p_k^2]$ are exactly the primes in I_k, and p_{k+1}^2 is the smallest composite orbit minimum ($289, 361, 529$ for $k = 6, 7, 8$).

Proof. Orbit minima are units, so by Lemma 4.3 the left side is contained in the right side. Conversely every element n of the right side is a unit with $n \leq p_{k+1}^2 \leq 2\sqrt{M_k}$ by Lemma 4.4, hence $n^2 \leq 4M_k$, and n is an orbit minimum by Lemma 4.2. Since $p_k^2 < p_{k+1}^2$, the window statement follows. □

Proposition 4.6 (Sharpness, **[S]**). *The window statement fails for $k = 3, 4, 5$. At $k = 5$ ($M_5 = 2310$, $2\sqrt{M_5} \approx 96.1$) all primes in $(11, 96]$ are orbit minima, but the four primes $97, 101, 103, 107$ are not: they lie in the orbits of $13, 31, 37, 47$, respectively, with $n^2 - z^2 = 4M_5$ in each case (e.g. $97^2 - 13^2 = 9240$). At $k = 4$, $31^2 - 11^2 = 840 = 4M_4$; at $k = 3$, $B_3 = \{1, 7\}$. The smallest composite orbit minimum at $k = 5$ is $247 = 13 \cdot 19$.*

Proof. Direct verification with the orbit criterion; the code is in Appendix B. □

Remark 4.7 (what completeness does and does not say). Theorem 4.5 is a *characterisation*, not an existence theorem for primes: it says that the orbit-minimum construction, applied in its window, produces the primes and nothing else. It does not by itself imply that the window contains a prime; that follows from Bertrand's postulate (Section 6). At $k = 6$ there are 27 composite orbit minima below $M_6/2$; above p_{k+1}^2 composites appear and their share grows with n.

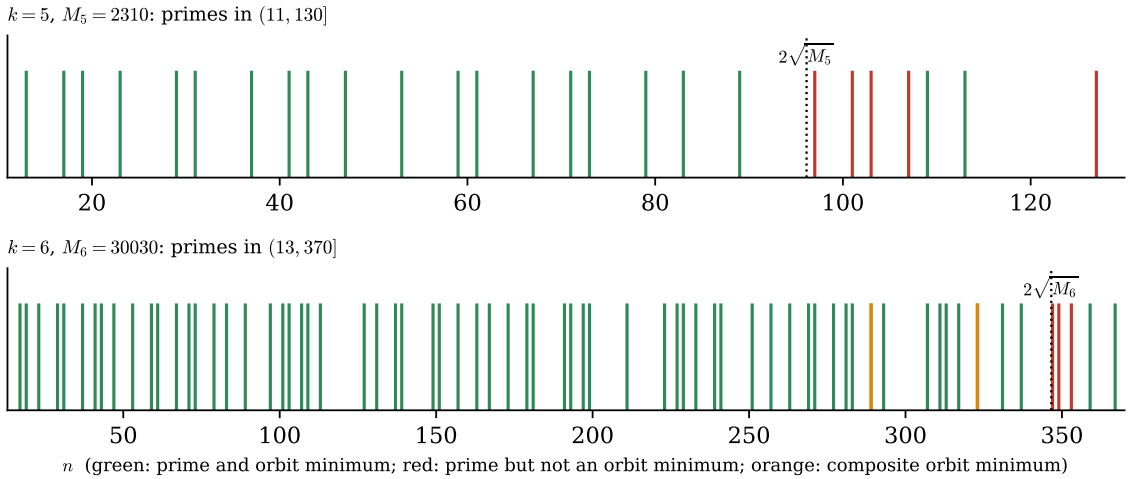


Figure 8: The threshold $2\sqrt{M_k}$ of the key lemma. Top ($k = 5$): primes up to $2\sqrt{M_5} \approx 96.1$ are orbit minima (green), the four primes 97–107 are not (red). Bottom ($k = 6$): every prime up to $2\sqrt{M_6} \approx 346.6$ is an orbit minimum; the first composite orbit minimum is $17^2 = 289$ (orange), as predicted by Theorem 4.5.

4.3 Self-similarity and recursion

Theorem 4.8 (Recursion, **[S]**). *1. For every $k \geq 3$, $p_{k+1} = \min(B_k \setminus \{1\})$.*

2. Reduction modulo M_k maps every G_{k+1}-orbit onto a G_k-orbit, and over every G_k-orbit lie exactly $(p_{k+1} - 1)/2$ orbits of level $k + 1$. Hence $|B_{k+1}| = |B_k|\,(p_{k+1} - 1)/2$.

Proof. (1) p_{k+1} is the smallest unit > 1. By Bertrand $p_{k+1} < 2p_k$, and $p_k^2 \leq M_k$ for $k \geq 3$ (since $M_{k-1} \geq 2p_{k-1} > p_k$), so $p_{k+1}^2 < 4p_k^2 \leq 4M_k$, and p_{k+1} is an orbit minimum by Lemma 4.2. No unit strictly between 1 and p_{k+1} exists. (2) By the Chinese remainder theorem the square map factorises: $y^2 \bmod M_{k+1} \leftrightarrow (y^2 \bmod M_k,\ y^2 \bmod p_{k+1})$. Orbits of level $k+1$ are therefore pairs (orbit of level k, non-zero square class modulo p_{k+1}), and $U(p_{k+1})$ has $(p_{k+1} - 1)/2$ squares. □

The passage $k \to k + 1$ is the *tick* of the machine [36]: *step* (multiply the modulus by p_{k+1}), *filter* (remove the multiples of p_{k+1}, so $|U|$ is multiplied by $p_{k+1} - 1$) and *mirror* (adjoin the new sign flip σ_{k+1}, so $|G|$ doubles).

4.4 Convergence and branching

Proposition 4.9 (**[S]**). *The vertex density of level k is $\varphi(M_k)/M_k = \prod_{p \leq p_k}(1 - 1/p) \sim e^{-\gamma}/\log p_k$ (Mertens), the branching factor of vertices is $p_{k+1} - 1$, and that of orbits is $(p_{k+1} - 1)/2$.*

The density is the probability that a random integer survives the sieve up to p_k. Near $x = p_k^2$ the survivors of the sieve up to $\sqrt{x}$ are exactly the primes, yet the sieve density there is $2e^{-\gamma}/\log x \approx 1.123/\log x$, not the prime density $1/\log x$. This classical mismatch (Mertens' constant $e^{-\gamma}$ instead of $1/2$) shows that local sieve counts do not automatically become prime counts; it is the simplest instance of the gap between steps 4 and 5 of the scheme in Section 5.

4.5 Cross-check in four languages

In plain words: four ways to see the same object

The same orbit structure can be described in four mathematical languages: as a graph (each orbit is a hypercube), as a linear operator (an averaging operator with a simple spectrum), as a table of quadratic residues (Legendre symbols), and as a growth law. They all say the same thing, which is a useful consistency check and makes the construction accessible from several fields [46].

Proposition 4.10 (**[S]**). *1.* Graph. *The Cayley graph of the action (generators $\sigma_2, \dots, \sigma_k$) is a disjoint union of $|B_k|$ copies of the hypercube Q_{k-1}.*

2. Operator. *Let U_j be the permutation operator of σ_j on functions $U(M_k) \to \mathbb{C}$ and $T = \frac{1}{k}(I + \sum_{j=2}^{k} U_j)$. The eigenvalues of T are $(k-2m)/k$, $m = 0, \dots, k-1$, with multiplicity $|B_k|\binom{k-1}{m}$.*

3. Legendre. *For an odd prime $p_i \neq p_j$ the Legendre symbol $(\sigma_j a/p_i) = (a/p_i)$, while $(\sigma_j a/p_j) = (-1/p_j)(a/p_j)$; thus σ_j flips the quadratic character modulo p_j if and only if $p_j \equiv 3 \pmod 4$.*

4. Growth. $|B_{k+1}|/|B_k| = (p_{k+1} - 1)/2$ *(Theorem 4.8).*

Proof. (1) A free action of $(\mathbb{Z}/2)^{k-1}$ with its standard generators makes each orbit a copy of the Cayley graph of $(\mathbb{Z}/2)^{k-1}$, which is Q_{k-1}. (2) On each orbit T acts as $\frac{1}{k}(1+\sum_j g_j)$ in the regular representation of $(\mathbb{Z}/2)^{k-1}$; on the character χ_S ($S \subseteq \{2, \dots, k\}$, $|S| = m$) it has the eigenvalue $\frac{1}{k}(1+(k-1-m)-m)$. (3) $\sigma_j a \equiv a \pmod{p_i}$ and $\equiv -a \pmod{p_j}$, and $(-1/p) = (-1)^{(p-1)/2}$. □

Numerical Observation 4.11 (non-locality, **[N]**). Call a unit a *local minimum* if it is smaller than each of its $k-1$ single-flip neighbours $\sigma_j(a)$. Every orbit minimum is a local minimum, but the converse fails from $k = 4$ on:

k	3	4	5	6	7	8
local minima	2	18	130	1263	18328	293743
orbit minima $\lvert B_k\rvert$	2	6	30	180	1440	12960

Being an orbit minimum is therefore a genuinely global property of the hypercube Q_{k-1}; it cannot be tested by looking at neighbours only. This is the first appearance of the non-locality that limits the method (Section 12).

4.6 Algorithmic position

Completeness yields an algorithm that lists the primes in (p_k, p_{k+1}^2) in increasing order as orbit minima [37]. Its cost is that of a wheel sieve with wheel M_k; it is a transparent reference implementation, not a competitor of optimised segmented sieves such as `primesieve` **[N]**. We use it only as an independent check of the theory.

5 The application scheme

In plain words

All classical questions about primes in patterns (twins, Goldbach pairs, prime triplets, ...) are treated by the same recipe. First, translate the pattern into a configuration on the polytope: twin primes are pairs of corners that differ by 2, Goldbach pairs are corners that add up to N. Second, determine on which half-axes such a configuration can live at all. Third, count exactly how many admissible configurations exist at level k; this is pure counting with the Chinese remainder theorem. Fourth, turn the count into a density; as k grows, it converges to the famous constant in the Hardy–Littlewood prediction. Fifth comes the step the geometry cannot do: show that the admissible configurations are really occupied by primes, infinitely often. We call this the *lifting problem*; for each problem we state what is known classically. Sixth, we compare the predictions with counts up to 10^8.

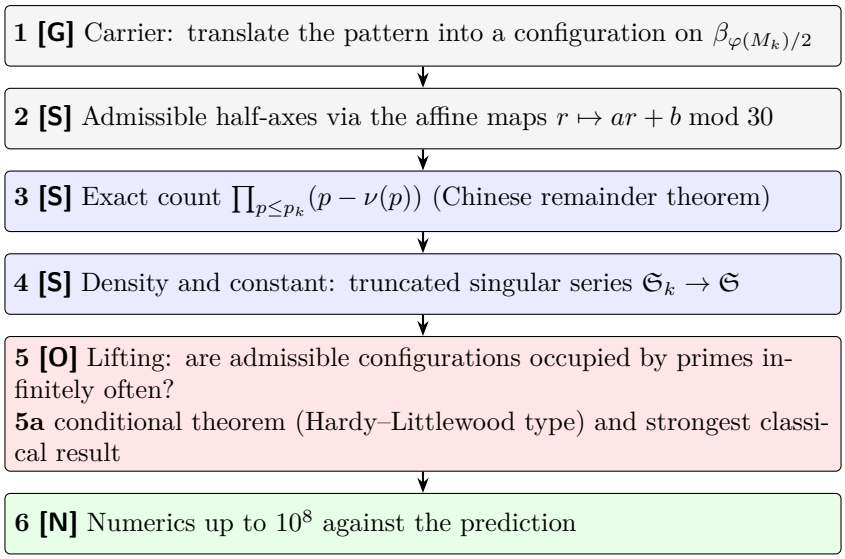


Figure 9: The application scheme. Steps 1–4 are exact; step 5 is where the classical difficulty sits; step 6 is the empirical check.

5.1 Admissible patterns and the exact count

A finite set $H = \{h_1, \dots, h_m\} \subset \mathbb{Z}$ (a *pattern*) is *admissible* if for every prime p the set $H \bmod p$ misses at least one residue class. Write $\nu_H(p) = \#(H \bmod p)$.

Proposition 5.1 (Exact count, **[S]**). *The number of $a \in \mathbb{Z}/M_k$ such that $a + h$ is a unit for every $h \in H$ is*

$$N_k(H) = \prod_{p \le p_k} (p - \nu_H(p)).$$

Proof. By the Chinese remainder theorem the condition splits into independent conditions modulo each $p \le p_k$, and modulo p the admissible a are those avoiding the $\nu_H(p)$ classes $-h \bmod p$. □

Proposition 5.2 (Truncated singular series, **[S]**). *Define $\mathfrak{S}_k(H) = N_k(H)\, M_k^{m-1}/\varphi(M_k)^m$. Then*

$$\mathfrak{S}_k(H) = \prod_{p \le p_k} \Big(1 - \frac{\nu_H(p)}{p}\Big)\Big(1 - \frac{1}{p}\Big)^{-m} \xrightarrow[k\to\infty]{} \mathfrak{S}(H),$$

the Hardy–Littlewood singular series of H [10], and the product converges absolutely. $\mathfrak{S}(H) > 0$ if and only if H is admissible.

Proof. The identity is Proposition 5.1 divided by $M_k \prod_p (1 - 1/p)^m$. For p larger than all differences $h_i - h_j$, $\nu_H(p) = m$ and the factor is $1 - \binom{m}{2}p^{-2} + O(p^{-3})$, so the product converges absolutely; it vanishes iff some factor vanishes, i.e. iff $\nu_H(p) = p$ for some p. □

Geometric Reading 5.3 (**[G]**). $N_k(H)$ is the number of placements of the pattern H on the vertices of $\beta_{\varphi(M_k)/2}$, and $\mathfrak{S}_k(H)$ compares it with m independent random vertices. The Hardy–Littlewood constant is thus the limit of an exact vertex count on the cascade.

5.2 Three classes of prime families

The atlas of prime families [51] sorts the classical families into three heuristic growth classes **[H]**: *Class 1a* (pairs and tuples with a polynomial relation: twins, cousins, sexy primes, Sophie Germain primes): count $\sim c\, x/\log^m x$ with c a singular series; *AP-factor class* (Mersenne, Fermat, repunit, Cullen, Woodall, Proth): divisibility forces every prime factor onto a progression

of small density, and the expected count grows like $c \log x$; *Class 3* (self-correlation conditions modulo p^2: Wieferich, Wilson, Wall–Sun–Sun, Wolstenholme): expected count $\sim c \log\log x$ from the divergence of $\sum 1/p$. Sections 7 and 8 treat Class 1a, Section 9 the other two.

6 Applications I: windows and completeness

In plain words

The completeness theorem says that, in a window between a prime and the square of the next prime, the orbit minima are exactly the primes. Several old questions ask whether such windows, or smaller ones, always contain primes. For the window from n to $2n$ the answer is yes (Bertrand's postulate, proved in 1852). For the gaps between consecutive squares (Legendre, Brocard) the answer is not known. The geometry turns these questions into exact statements about how many orbit minima lie in a given range, but it does not supply the missing analytic estimate.

6.1 Bertrand

Theorem 6.1 (Chebyshev; Erdős' proof [7], **[S]**). *For every $n \geq 1$ there is a prime p with $n < p \leq 2n$.*

Erdős' proof compares the size of the central binomial coefficient $\binom{2n}{n} \geq 4^n/(2n+1)$ with the product of its prime-power factors: if $(n, 2n]$ contained no prime, every prime factor would be $\leq 2n/3$, which forces $\binom{2n}{n} \leq (2n)^{\sqrt{2n}}\, 4^{2n/3}$, impossible for large n; small n are checked directly. In [39] this argument is read as a balance on "Pascal's dome" **[G]**. In the cascade, Bertrand guarantees the ladder $p_k < p_{k+1} < 2p_k$, i.e. the step of every tick lands in the window I_k, and it is the only analytic input used in Lemma 4.4 and Theorem 4.8.

6.2 Brocard

Geometric Reading 6.2 (**[G]**). For $k \geq 6$, Theorem 4.5 gives

$$\#\{\text{primes in } (p_k^2, p_{k+1}^2)\} = \#(B_k \cap (p_k^2, p_{k+1}^2)).$$

Brocard's conjecture (at least four primes between p_k^2 and p_{k+1}^2 for $k \geq 2$) is therefore equivalent, for $k \geq 6$, to the statement that the upper part (p_k^2, p_{k+1}^2) of the completeness range always contains at least four orbit minima.

Numerical Observation 6.3 (**[N]**). For $k = 2, \ldots, 15$ the number of primes in (p_k^2, p_{k+1}^2) is $5, 6, 15, 9, 22, 11, 27, 47, 16, 57, 44, 20, 46, 80$. The conjecture is open **[O]**.

6.3 Legendre, Andrica, Oppermann

Legendre's conjecture (a prime between n^2 and $(n+1)^2$), Andrica's conjecture ($\sqrt{p_{n+1}} - \sqrt{p_n} < 1$) and Oppermann's conjecture (primes in $(n^2 - n, n^2)$ and $(n^2, n^2 + n)$) are all open **[O]**. Each would follow from a gap bound $p_{n+1} - p_n \ll p_n^{1/2-\varepsilon}$. The best unconditional exponent is 0.525 (Baker–Harman–Pintz [2]); between consecutive *cubes* there is a prime for all large n (Ingham [13]). In the geometry all three statements are window statements of the type of Geometric Reading 6.2 **[G]**; the method gives the exact count of candidates (units) in every window, but no lower bound for the primes among them.

6.4 Fermat numbers

Proposition 6.4 (**[S]**). *For $n \geq 2$ the Fermat number $F_n = 2^{2^n}+1$ satisfies $F_n \equiv 17 \pmod{30}$; it lies on the half-axis A_4^-. Every prime factor q of F_n satisfies $q \equiv 1 \pmod{2^{n+2}}$.*

Proof. $2^{2^n} \equiv 1 \pmod 3$ (even exponent) and $\pmod 5$ (exponent divisible by 4), so $F_n \equiv 2 \pmod{15}$, and F_n is odd. The divisor statement is the classical theorem of Euler and Lucas [11]. □

Since $\sum_n 1/\log F_n$ converges, the probabilistic heuristic predicts only finitely many Fermat primes **[H]** [11]; whether there is a Fermat prime beyond $F_4 = 65537$ is open **[O]**. In [43] the heuristic is read as a statement about the shrinking share of the progression $1 \bmod 2^{n+2}$ on the cascade **[G]**. Note the coincidence with Proposition 2.8: 30 is built only from Fermat primes, and every F_n ($n \geq 2$) sits on one fixed half-axis of its 16-cell.

7 Applications II: pairs (translation and antipode)

In plain words

There are two basic ways in which two corners of the polytope can be related. They can differ by a fixed amount (a *translation*): twin primes differ by 2, "sexy" primes by 6. Or they can add up to a fixed amount (an *antipode*, generalised): a Goldbach pair adds up to N. In both cases the polytope tells us exactly on which corners such pairs can sit and how many admissible pairs there are at each level. The count explains, for instance, why even numbers divisible by 30 have about 8/3 times as many Goldbach representations as numbers like $2 \cdot 7$: the factor 8/3 is the number of the ratio 8 : 3 of the ordered pair counts for axes and for edges plus diagonal in Theorem 2.4.

7.1 Twin primes

Theorem 7.1 (**[S]**). *1. The lower member $p > 5$ of a twin pair lies in one of the classes $\{11, 17, 29\} \pmod{30}$; the pair $\{p, p+2\}$ is always an edge (never an axis) of every $\beta_{\varphi(M_k)/2}$.*

2. The number of twin configurations at level k is $N_k(\{0,2\}) = \prod_{3 \leq p \leq p_k}(p-2)$, i.e. 1, 3, 15, 135, 1485, 22275 for $k = 2, \ldots, 7$.

3. The renormalised edge density $\delta_k = N_k(\{0,2\})/\varphi(M_k)$ satisfies

$$\frac{\delta_k \log p_k}{2e^{-\gamma}} \longrightarrow C_2 = \prod_{p \geq 3}\Big(1 - \frac{1}{(p-1)^2}\Big) = 0.6601618\ldots$$

Proof. (1) r and $r+2$ must both be units modulo 30. A pair $\{a, a+2\}$ is an axis iff $2a + 2 \equiv 0 \pmod{M_k}$, i.e. $a \equiv M_k/2 - 1$, which is even and hence not a unit. (2) is Proposition 5.1. (3) $(p-2)/(p-1) = \frac{p(p-2)}{(p-1)^2} \cdot \frac{p-1}{p}$, hence $\delta_k = C_2^{(k)} \cdot 2\prod_{p \leq p_k}(1 - 1/p)$ with the partial product $C_2^{(k)} \to C_2$, and Mertens' theorem gives the limit. □

The convergence in (3) is slow but monotone in tendency: the renormalised density is 0.5375 at $k = 3$, 0.5889 at $k = 6$, 0.6301 at $k = 10$ and 0.6497 at $k = 39$ **[N]** (Figure 10); see [52, 42].

Conditional Theorem 7.2 (Hardy–Littlewood [10]). *If the prime pair conjecture holds for $H = \{0, 2\}$, then $\pi_2(x) = \#\{p \leq x : p+2 \text{ prime}\} \sim 2C_2\,\mathrm{Li}_2(x)$, with $\mathrm{Li}_m(x) = \int_2^x \mathrm{d}t/\log^m t$.*

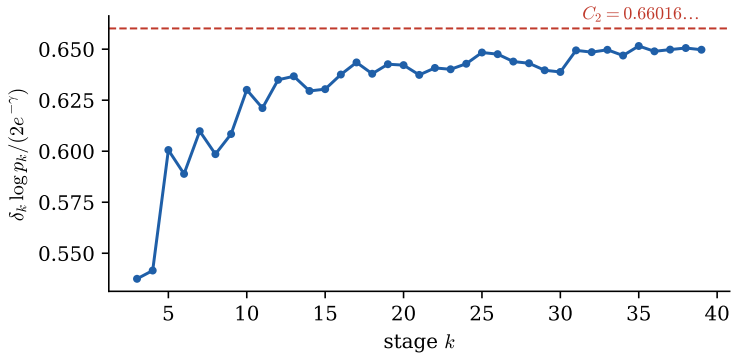


Figure 10: Renormalised twin-edge density $\delta_k \log p_k/(2e^{-\gamma})$ for $k = 3, \ldots, 39$ and its limit C_2 (Theorem 7.1).

Classical anchor. Infinitely many primes p with $p+2$ having at most two prime factors (Chen [4]); infinitely many prime gaps ≤ 246 (Zhang [30], Maynard [17], Polymath [23]). In the language of the cascade: *some* admissible edge pattern of length ≤ 246 lifts infinitely often; for the twin edge itself the lifting problem is open **[O]**.

Numerical Observation 7.3 ([N]).

x	$\pi_2(x)$	$2C_2 \operatorname{Li}_2(x)$	ratio
10^4	205	214	0.957
10^5	1 224	1 249	0.980
10^6	8 169	8 248	0.990
10^7	58 980	58 754	1.004
10^8	440 312	440 368	0.9999

7.2 Polignac

For even d the pattern $\{0, d\}$ is admissible, and by Proposition 5.1

$$N_k(\{0,d\}) = \prod_{3\leq p\leq p_k,\ p\mid d} (p-1) \prod_{3\leq p\leq p_k,\ p\nmid d} (p-2), \qquad \frac{\mathfrak{S}(\{0,d\})}{\mathfrak{S}(\{0,2\})} = \prod_{p\mid d,\ p>2} \frac{p-1}{p-2} \quad \textbf{[S]}.$$

So $d = 6$ should have twice as many prime pairs as $d = 2$, and $d = 30$ should have $2 \cdot \frac{4}{3} = \frac{8}{3}$ times as many **[H]**. Geometrically, $p \mid d$ means that the translation by d fixes the p-coordinate, so no vertex is lost modulo p **[G]** [44]. Up to 10^8 the measured ratios are 1.998 ($d = 6$) and 2.666 ($d = 30$) **[N]**. Polignac's conjecture (every even d occurs infinitely often) is open **[O]**; the Maynard–Polymath theorem shows that at least one $d \leq 246$ does.

7.3 Goldbach

Theorem 7.4 (Local Goldbach count, **[S]**). *For even N the number of ordered pairs $(a, b) \in U(M_k)^2$ with $a + b \equiv N \pmod{M_k}$ is*

$$g_k(N) = \prod_{3\leq p\leq p_k} N_p(N), \qquad N_p(N) = \begin{cases} p-1, & p \mid N, \\ p-2, & p \nmid N. \end{cases}$$

For $N \equiv 0 \pmod{M_k}$ the pairs are exactly the axes (antipodes); in general they are the orbits of the involution $a \mapsto N - a$, a translated antipode.

Proof. As in Theorem 2.4, prime by prime. □

The ratio $g_k(N)/g_k(2)$ is $\prod_{p\mid N,\, p\le p_k} \frac{p-1}{p-2}$: at $k=3$ the four values 8/3, 2, 4/3, 1 for $30 \mid N$, $6 \mid N$, $10 \mid N$ and otherwise are exactly the ratios of axes, edges and diagonals in Theorem 2.4.

Conditional Theorem 7.5 (Hardy–Littlewood [10])**.** *If the Hardy–Littlewood conjecture for the binary Goldbach problem holds, then the number of ordered representations satisfies*

$$r(N) = \#\{(p,q) : p+q = N\} \sim 2C_2 \prod_{p\mid N,\, p>2} \frac{p-1}{p-2} \frac{N}{\log^2 N}.$$

Classical anchor. Every large even N is $p + P_2$ (Chen [4]); the exceptional set of even $N \le x$ that are not sums of two primes is $O(x^{1-\delta})$ (Montgomery–Vaughan [19]); every odd $N \ge 7$ is a sum of three primes (Helfgott [12]); the binary conjecture is verified up to $4 \cdot 10^{18}$ [22]. The binary conjecture itself is open **[O]**. In [40, 45] the conjecture is formulated as an antipode axiom on the cascade **[G]**; such an axiom is equivalent to the conjecture and does not prove it.

Numerical Observation 7.6 (**[N]**)**.** For even $N \in (20000, 40000]$ the mean of $r(N)$ is 1303 ($30 \mid N$), 978 ($6 \mid N$, $30 \nmid N$), 652 ($10 \mid N$, $30 \nmid N$) and 490 (otherwise). The ratios 2.66, 2.00, 1.33 match 8/3, 2, 4/3 (Figure 11).

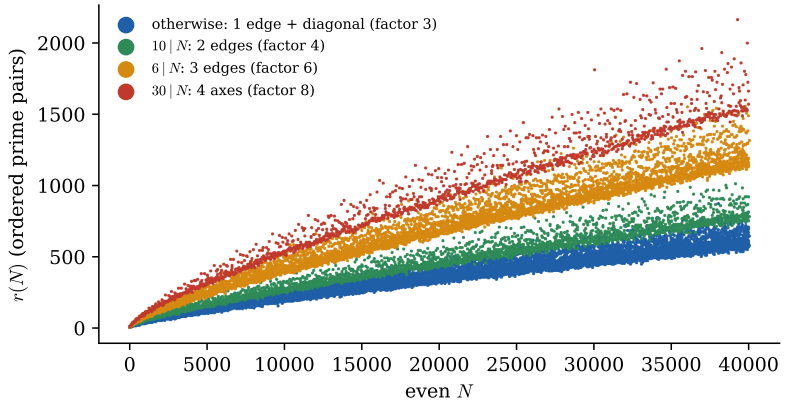


Figure 11: The Goldbach comet $r(N)$ for even $N \le 40000$, coloured by the four classes of the local factor at 30 (Theorem 2.4). The bands are ordered by the factors $3, 4, 6, 8$; further splitting within the bands comes from the primes $7, 11, \ldots$ dividing N.

7.4 Sophie Germain primes and Cunningham chains

Proposition 7.7 (**[S]**)**.** *If $p > 5$ and $2p+1$ are both prime, then $p \equiv 11, 23, 29 \pmod{30}$, and the map $p \mapsto 2p+1$ sends these classes to $23, 17, 29$. Consequently, in a Cunningham chain $p, 2p+1, 4p+3, \ldots$ of length ≥ 4 with all members > 5, every member is $\equiv 29 \pmod{30}$, i.e. the whole chain sits on the single vertex A_1^- of the 16-cell.*

Proof. $2r+1$ is a unit modulo 30 exactly for $r \in \{11, 23, 29\}$ among the units. Starting in 11 the chain runs $11 \to 23 \to 17$ and stops (as 17 is not admissible), starting in 23 it runs $23 \to 17$; only $29 \mapsto 29$ can be iterated. □

Under the Bateman–Horn conjecture [3] the number of Sophie Germain primes up to x is $\sim 2C_2\, x/(\log x \log 2x)$ **[H]**; infinitude is open **[O]**. Example: the chain $89, 179, 359, 719, 1439, 2879$ lies on A_1^-.

7.5 Chen and Lemoine

Lemoine's conjecture (every odd $N \geq 7$ is $p + 2q$ with primes p, q) has the same local structure as Goldbach: the number of $(a, b) \in U(M_k)^2$ with $a + 2b \equiv N$ is again $\prod_{3 \leq p \leq p_k} N_p(N)$ **[S]**, because $b \mapsto 2b$ is a bijection of $U(p)$. Chen's theorem is the strongest classical anchor for both families; the lifting problems are open **[O]**.

7.6 Prime races

By Dirichlet's theorem and the prime number theorem in progressions, the eight vertices of the 16-cell are asymptotically equally occupied [6]: up to 10^8 each class contains about 720 000 primes **[N]**. Finer effects are not equidistributed. Chebyshev's bias (non-residue classes lead most of the time) is explained, under GRH and a linear independence hypothesis, by Rubinstein–Sarnak [26]. Consecutive primes avoid repeating their residue class (Lemke Oliver–Soundararajan [16]): up to 10^8 the proportion of consecutive primes in the *same* class modulo 30 is 0.0443, against $1/8 = 0.125$ for independent classes **[N]**. In the geometry, a step to the same vertex is the rarest step; the explanation is analytic (second-order terms of the Hardy–Littlewood conjecture), not geometric.

8 Applications III: tuples and polynomials

In plain words

The same recipe works for longer patterns: prime triplets such as $(p, p+2, p+6)$, quadruplets $(p, p+2, p+6, p+8)$, and values of polynomials such as $n^2 + 1$. In each case the polytope picks out the admissible corners (for $n^2 + 1$ they form a single tetrahedral cell of the 16-cell), the exact count converges to the Hardy–Littlewood constant, and the numbers up to 10^8 match the prediction to within a few parts per thousand. For long arithmetic progressions of primes the situation is different: here the existence problem has been solved (Green–Tao), and the geometry shows why such progressions must sit on a single corner.

8.1 Hardy–Littlewood constellations

For an admissible H with m elements, Propositions 5.1 and 5.2 give the exact count and the constant. The Hardy–Littlewood conjecture [10] predicts $\#\{p \leq x : p + h \text{ prime } \forall h \in H\} \sim \mathfrak{S}(H)\,\mathrm{Li}_m(x)$; it is open for every $m \geq 2$ **[O]**. The Maynard–Tao method [17] shows: for every m, every sufficiently large admissible pattern contains m elements that are simultaneously prime for infinitely many translates.

Numerical Observation 8.1 (**[N]**)**.** Up to $x = 10^8$: triplets $\{0, 2, 6\}$: 55 600; $\{0, 4, 6\}$: 55 556; prediction $\mathfrak{S}\,\mathrm{Li}_3(x) = 55\,491$ with $\mathfrak{S} = 2.8582$. Quadruplets $\{0, 2, 6, 8\}$: 4 768; prediction 4 735 with $\mathfrak{S} = 4.1512$.

Exact counts $N_k(H)$ for these patterns at $k = 2, \ldots, 7$ are in Appendix C; see [41].

8.2 Landau $n^2 + 1$ and Bunyakovsky–Schinzel

Theorem 8.2 (**[S]**)**.** *Every prime of the form $n^2 + 1$ larger than 5 is congruent to 1, 7, 11 or 17 modulo 30. These four classes form one tetrahedral cell of the 16-cell (one vertex on each axis: $A_1^+, A_2^+, A_3^+, A_4^-$).*

Proof. n must be even, and running through the even residues modulo 30 gives $n^2 + 1 \bmod 30 \in \{1, 5, 7, 11, 17, 25\}$; the classes 5 and 25 are excluded. The four remaining classes lie on four different axes, hence span a cell. □

Conditional Theorem 8.3 (Hardy–Littlewood conjecture F [10], Bateman–Horn [3]). *$\#\{n \le x : n^2+1$ prime$\} \sim \frac{C}{2}\,\mathrm{li}(x)$ with $C = \prod_{p>2}(1 - \frac{(-1/p)}{p-1}) = 1.37281\ldots$*

Numerical Observation 8.4 (**[N]**). For $x = 10^3, 10^4, 10^5, 2\cdot 10^5$ the counts are 112, 841, 6 656, 12 391; the predictions are 121, 855, 6 609, 12 379.

Classical anchor. n^2+1 has at most two prime factors for infinitely many n (Iwaniec [14]). Landau's problem is open **[O]**. The cell of Theorem 8.2 is the geometric carrier; the constant C is the limit of its truncated singular series, where the Legendre symbol $(-1/p)$ (Proposition 4.10(3)) decides whether p removes two or zero residues.

8.3 Primes in arithmetic progression (Green–Tao)

Proposition 8.5 (**[S]**). *Let $a, a+d, \ldots, a+(\ell-1)d$ be primes, all $> \ell$. Then every prime $q \le \ell$ divides d. Hence, if $p_k \le \ell$, all terms are congruent modulo M_k: the whole progression sits on a single vertex of $\beta_{\varphi(M_k)/2}$.*

Proof. If $q \nmid d$, the $\ell \ge q$ terms cover all residues modulo q, so one term is divisible by q; being $> \ell \ge q$, it is not prime. □

Green and Tao proved that the primes contain arbitrarily long arithmetic progressions [9]. In the language of the cascade: the "single-vertex" configurations of every length lift **[S]**. This is one of the few classical lifting problems that is solved, and it shows that the lifting step requires analytic input of a completely different kind (higher-order Fourier analysis, transference) from the local counts that the geometry provides.

9 Applications IV: exponential families and self-correlation

In plain words

Some famous prime families grow exponentially: Mersenne numbers 2^p-1, Fermat numbers $2^{2^n}+1$, repunits $11\ldots1$. For them the polytope does something simple and exact: it tells us on which corner of the 16-cell the members must sit (Mersenne primes on two corners, repunit primes on two others). The expected number of such primes grows only like a logarithm. A second group (Wieferich, Wilson, Wolstenholme primes) is defined by a condition modulo p^2 at the prime itself; such conditions are invisible to the polytope, which only sees remainders modulo small fixed primes. Finally, there are numbers k for which $k\cdot 2^n+1$ is *never* prime; here the local obstruction is total, and the polytope sees it completely.

9.1 AP-factor families

Proposition 9.1 (**[S]**). *Let $p \ge 5$ be prime. Then $2^p - 1 \equiv 1 \pmod{30}$ if $p \equiv 1 \pmod 4$ and $2^p - 1 \equiv 7 \pmod{30}$ if $p \equiv 3 \pmod 4$. Every prime factor q of 2^p-1 satisfies $q \equiv 1 \pmod{2p}$ and $q \equiv \pm 1 \pmod 8$.*

Proof. $2^p \equiv 2 \pmod 3$ for odd p, and $2^p \bmod 5$ is 2 or 3 according as $p \equiv 1$ or $3 \pmod 4$; combine with oddness. The divisor statement is classical: the order of 2 modulo q is p, so $p \mid q-1$; and 2 is a quadratic residue modulo q since $2 \equiv (2^{(p+1)/2})^2$ [11]. □

So Mersenne primes beyond 7 occupy only the vertices A_1^+ and A_2^+. The Lenstra–Pomerance–Wagstaff heuristic predicts $\#\{p \le x : 2^p-1 \text{ prime}\} \sim (e^\gamma/\log 2)\log x$ **[H]** [29].

Proposition 9.2 (**[S]**). *Let $R_n = (10^n-1)/9$ be the repunit with $n > 3$ prime. Then $R_n \equiv 1 \pmod{30}$ if $n \equiv 1 \pmod 3$ and $R_n \equiv 11 \pmod{30}$ if $n \equiv 2 \pmod 3$.*

Proof. R_n is odd and $\equiv 1 \pmod 5$; modulo 3 it is congruent to its digit sum n. □

Cullen ($n2^n+1$), Woodall ($n2^n-1$) and Proth-type families ($k2^n+1$ for fixed k) are handled in the same way: a half-axis or a small set of half-axes is forced by the exponent modulo small numbers, and the heuristic count grows logarithmically **[H]** [51].

9.2 Sierpiński and Riesel

For $k = 78557$ every number $k \cdot 2^n + 1$ is divisible by one of $3, 5, 7, 13, 19, 37, 73$ (Selfridge, see [27]); for $k = 509203$ every $k \cdot 2^n - 1$ is divisible by one of $3, 5, 7, 13, 17, 241$ (Riesel [25]) **[S]**. In the language of the cascade, a covering congruence means that for each exponent n the number lands on a *non-unit* residue at some fixed finite level: the local obstruction is complete, and the count of Proposition 5.1 is zero. These families are the boundary case of the growth law: the logarithmic heuristic has constant 0.

9.3 Self-correlation families

Wieferich primes ($2^{p-1} \equiv 1 \bmod p^2$; known: 1093, 3511), Wilson primes ($(p-1)! \equiv -1 \bmod p^2$; known: 5, 13, 563), Wall–Sun–Sun primes (none known) and Wolstenholme primes ($\binom{2p-1}{p-1} \equiv 1 \bmod p^4$; known: 16843, 2124679) are defined by a condition modulo a power of the prime itself **[N]**. If such a condition is satisfied with probability $1/p$, the expected number up to x is $\sum_{p \le x} 1/p \sim \log\log x$ **[H]**. The condition is not a residue condition modulo the fixed small primes of the cascade; the geometry is blind to these families **[G]**, which is an honest limit of the method.

9.4 Boundary objects and fractal classification

Fibonacci and Lucas primes reduce to progressions through the rank of apparition; Carmichael numbers (Korselt's criterion) are composites that pass the Fermat test, and there are infinitely many [1]. In [51] prime families are also compared by the finite-scale box-counting dimension of their logarithmic embedding $p \mapsto \log p / \log p_{\max} \in [0, 1]$: positive density families sit near 0.87, constellations lower, and exponential families tend to 0 (a "log-ladder") **[N]**. This is a diagnostic, not a topological invariant (every countable set has Hausdorff dimension 0).

10 Collatz as a test case for non-locality

In plain words

The Collatz rule ($n \mapsto 3n+1$ for odd n, $n \mapsto n/2$ for even n) is not a question about primes, but it is a perfect test of the method. Its statistics look exactly like a random process with a downward drift, and all local information (remainders modulo small numbers) is evenly spread. Yet nobody can prove that every starting value reaches 1. The case shows in pure form what the whole paper finds: local data can explain averages, but not the fate of every single number.

Cycle equation [S]. Write an orbit through odd numbers $x_1, \ldots, x_n$ with $x_{i+1} = (3x_i+1)/2^{a_i}$. A cycle of length n with $m = \sum a_i$ halvings satisfies $2^m = \prod_{i=1}^{n}(3 + 1/x_i)$, so m/n is close to $\log_2 3$ for cycles with large elements; this is the starting point of all known lower bounds on cycle lengths [15].

Drift [H]. If the number $a_i = v_2(3x_i+1)$ of halvings after an odd step behaves like a geometric random variable with mean 2, the mean change of $\log_2 x$ per odd step is $\log_2 3 - 2 \approx -0.415 < 0$. For the generalised map $n \mapsto pn + 1$ the drift is $\log_2 p - 2$, negative only for $p = 3$ among odd primes [32].

Numerics [N]. The value 9232 is the maximum of the trajectory for exactly 1579 starting values (all of them below 9232, so the count is the same for $N = 10^4, 10^5, 10^6$). The detrended log-trajectories show a spectral crossover near period six, twice the mean distance between odd steps [32].

Anchor and limit. Tao proved that almost all orbits attain almost bounded values [28]; the conjecture is open **[O]**. The map $n \mapsto 3n + 1$ sends every odd n to the class 1 mod 3, so it does not preserve the unit structure of the cascade; there is no orbit-minimum formulation. Collatz is included as a control case for non-locality (Section 12).

11 Spectral reflection

In plain words

The constants that appeared above (the twin constant C_2, Artin's constant, the Landau–Ramanujan constant, the Mertens constant) can all be written as the "determinant" of a very simple operator: one number per prime, placed on a diagonal. This repackaging is exact and elementary. Its use is that it shows which constants belong together. In particular the split of the primes by their remainder modulo 4 appears three times in this paper (sums of two squares, the Legendre sign of the flips, and the Landau–Ramanujan constant), and the operator picture makes it one and the same projection.

Let D be the diagonal operator on $\ell^2(\{\text{primes}\})$ and t_p its entries. If $\sum_p |t_p| < \infty$, then $\exp(\operatorname{tr} D) = \prod_p e^{t_p}$ is a Fredholm-type determinant **[S]**. With

constant	e^{t_p}	value
C_2 (twins)	$1 - (p-1)^{-2}$, $p \geq 3$	$0.6601618\ldots$
Artin A	$1 - \frac{1}{p(p-1)}$	$0.3739558\ldots$
$e^{M-\gamma}$ (Mertens)	$(1 - \frac{1}{p})e^{1/p}$	$e^{-0.3157\ldots}$
$\sqrt{2}\,K$ (Landau–Ramanujan)	$(1 - p^{-2})^{-1/2}$ for $p \equiv 3 \bmod 4$, else 1	$\sqrt{2} \cdot 0.7642236\ldots$

the four constants are exponentials of traces [52, 53]. The last line uses the projector P_3 onto the primes $p \equiv 3 \pmod 4$; the same projector selects the primes whose sign flip changes the quadratic character (Proposition 4.10(3)) and the primes that are not sums of two squares. This is a bookkeeping identity **[G]**; it carries no spectral information about zeros of ζ.

12 Two notions of closure and the limit of the method

In plain words

Two different things can be called "closing" a problem about primes. The geometric closure is complete: for every pattern we know exactly which corners are allowed and how many there are, at every level, and how this converges to the predicted constant. The analytic closure is the statement that the allowed corners are really filled with primes, again and again. Only the second one decides the famous conjectures, and it does not follow from the first. The reason is a kind of non-locality: whether a number is a prime depends on all of its potential divisors at once, not on a fixed finite set of them; the orbit-minimum property is already global on each hypercube (Numerical Observation 4.11).

Geometric closure [S]. For every admissible pattern the exact count $N_k(H)$, its truncated singular series and the limit $\mathfrak{S}(H)$ are determined (Propositions 5.1 and 5.2); the same holds for the Goldbach and Lemoine counts (Theorem 7.4) and for the polynomial case $n^2 + 1$ (Theorem 8.2).

Analytic closure. The step from local counts to primes requires estimates for primes in short intervals or progressions beyond what sieve methods give. A classical reason is the parity barrier of the Selberg sieve: local information cannot distinguish numbers with an even from numbers with an odd number of prime factors [8]. This is why Chen's theorems stop at P_2, and why the cascade, which consists of local information only, cannot replace the analytic step.

Table 1: Status of the lifting problem in the language of the cascade.

configuration	lifting status	strongest classical result
single vertex (APs mod q)	solved **[S]**	Dirichlet; PNT in progressions
single vertex, long APs	solved **[S]**	Green–Tao [9]
some edge of length ≤ 246	solved **[S]**	Maynard, Polymath [17, 23]
translated antipode, 3 summands	solved **[S]**	Helfgott [12]
twin edge $\{0, 2\}$	open **[O]**	Chen [4]
Polignac edges $\{0, d\}$	open **[O]**	bounded gaps
antipode $p + q = N$ (Goldbach)	open **[O]**	Chen; Montgomery–Vaughan [19]
cell $n^2 + 1$	open **[O]**	Iwaniec [14]
windows (Legendre, Brocard)	open **[O]**	Baker–Harman–Pintz [2]
half-axis A_4^- (Fermat)	open **[O]**	heuristic finiteness
Collatz (control case)	open **[O]**	Tao [28]

13 Outlook

Three directions of the series lie outside the scope of this synthesis and are only mentioned. (i) The antipodal involution defines a "half-line" on which several classical halvings coincide [49]. (ii) Minima of an amplitude derived from the matchstick machine on the line $\Re s = \frac{1}{2}$ show a numerical coincidence with zeros of ζ [48]; this is a numerical observation, not an argument for the Riemann hypothesis. (iii) A generator model of primality from geometric principles is developed in [54]. The tower of fields above the cascade and the interpretative strand of the series are treated separately.

A Proofs in full

All proofs of Sections 2 to 5 are complete in the text. We collect here the two points where a reader may want more detail.

A.1 Lemma 4.4, the finite part. The induction step needs $p_{k+1} \geq 17$, i.e. $k \geq 6$, and the base case $4M_6 = 120120 \geq 83521 = 17^4$. For $k \leq 5$ the inequality fails, as listed. Hence the equivalence.

A.2 Orbit partners at $k = 5$. Using Proposition 3.2, n is not an orbit minimum iff there is a unit $z < n$ with $z^2 \equiv n^2 \pmod{2310}$. For $n = 97, 101, 103, 107$ the smallest such z is $13, 31, 37, 47$, and in each case $n^2 - z^2 = 9240 = 4M_5$, the smallest value allowed by Lemma 4.2. For the primes $n \leq 96$ no such z exists, because $n^2 < 4M_5 = 9240$.

A.3 Spectrum of T. Fix an orbit O and $a \in O$. The map $g \mapsto ga$ identifies O with G_k, and functions on O with the regular representation. The characters $\chi_S(g) = \prod_{j \in S} \epsilon_j(g)$ form an eigenbasis of every U_j, with $U_j \chi_S = \pm \chi_S$ according as $j \notin S$ or $j \in S$. Summing gives the eigenvalue $(1 + (k - 1 - |S|) - |S|)/k$.

B Reference code

The following Python code (requires `sympy`) computes orbit minima via the orbit criterion and checks Theorem 4.5, Proposition 4.6, and Corollary 3.4 for $k = 3, \dots, 7$; it also reproduces the incidence counts of Proposition 2.3 and the local minima of Numerical Observation 4.11.

```
from sympy import primorial, prime, isprime, primerange
from math import gcd
from itertools import combinations

def orbit_minima(k):
    """B_k: smallest representative of every square class of U(M_k)."""
    M = int(primorial(k)); best = {}
    for y in range(1, M):
        if gcd(y, M) == 1:
            s = y * y % M
            if s not in best: # y increases, so the first hit is the minimum
                best[s] = y
    return M, sorted(best.values())

def check(k):
    M, B = orbit_minima(k); pk, pn = prime(k), prime(k + 1)
    phi = 1
    for i in range(1, k + 1): phi *= prime(i) - 1
    low = [b for b in B if 1 < b <= pn * pn]
    pred = list(primerange(pk + 1, pn * pn)) + [pn * pn]
    comp = [b for b in B if b > 1 and not isprime(b)]
    return dict(k=k, orbits=len(B), formula=phi // 2**(k - 1),
                completeness=(low == pred), smallest_composite=min(comp) if comp else None,
                p_next=min(b for b in B if b > 1))

def local_minima(k):
    """Units smaller than all single-flip neighbours sigma_j(a)."""
    M = int(primorial(k)); E = []
    for j in range(2, k + 1):
        p = prime(j); m = M // p
        E.append(next(e for e in range(1, M) if e % p == p - 1 and e % m == 1 % m))
    return sum(1 for a in range(1, M) if gcd(a, M) == 1
               and all(a < e * a % M for e in E))

def incidences():
    U = [a for a in range(1, 30) if gcd(a, 30) == 1]
    face = lambda S: all((a + b) % 30 for a, b in combinations(S, 2))
    edges = [S for S in combinations(U, 2) if face(S)]
    facets = [S for S in combinations(U, 3) if face(S)]
    cells = [S for S in combinations(U, 4) if face(S)]
    return len(edges), len(facets), len(cells) # (24, 32, 16)

if __name__ == "__main__":
    for k in range(3, 8):
        print(check(k), local_minima(k) if k <= 6 else "")
    print(incidences())
```

C Tables

Table 2: Exact counts $N_k(H)$ of Proposition 5.1 for the twin pattern $\{0,2\}$, the Polignac pattern $\{0,6\}$, the triplet $\{0,2,6\}$ and the quadruplet $\{0,2,6,8\}$.

k	M_k	$\varphi(M_k)$	$\{0,2\}$	$\{0,6\}$	$\{0,2,6\}$	$\{0,2,6,8\}$
2	6	2	1	2	1	1
3	30	8	3	6	2	1
4	210	48	15	30	8	3
5	2 310	480	135	270	64	21
6	30 030	5 760	1 485	2 970	640	189
7	510 510	92 160	22 275	44 550	8 960	2 457

Table 3: Addresses $(k, A_i, \pm, s)$ of the primes $7 \le p \le 113$ (Definition 2.9).

p	address	p	address	p	address	p	address
7	$(3, A_2, +, 0)$	31	$(4, A_1, +, 1)$	59	$(4, A_1, -, 1)$	89	$(4, A_1, -, 2)$
11	$(3, A_3, +, 0)$	37	$(4, A_2, +, 0)$	61	$(4, A_1, +, 2)$	97	$(4, A_2, +, 2)$
13	$(3, A_4, +, 0)$	41	$(4, A_3, +, 1)$	67	$(4, A_2, +, 1)$	101	$(4, A_3, +, 3)$
17	$(3, A_4, -, 0)$	43	$(4, A_4, +, 1)$	71	$(4, A_3, +, 2)$	103	$(4, A_4, +, 3)$
19	$(3, A_3, -, 0)$	47	$(4, A_4, -, 1)$	73	$(4, A_4, +, 2)$	107	$(4, A_4, -, 2)$
23	$(3, A_2, -, 0)$	53	$(4, A_2, -, 1)$	79	$(4, A_3, -, 1)$	109	$(4, A_3, -, 2)$
29	$(3, A_1, -, 0)$			83	$(4, A_2, -, 2)$	113	$(4, A_2, -, 3)$

D Glossary for non-specialists

Unit modulo M. A remainder a with $\gcd(a, M) = 1$; only units can be remainders of primes not dividing M.

Primorial M_k. The product of the first k primes: $2, 6, 30, 210, 2310, \ldots$

Cross-polytope β_d. The d-dimensional generalisation of the octahedron: $2d$ corners $\pm e_i$, every pair of non-opposite corners joined. The 16-cell is β_4.

Axis, half-axis. A pair $\{a, M-a\}$ of mirror remainders; each of its two members is a half-axis.

Sign flip σ_j. Changing the sign of a remainder modulo one prime p_j only.

Orbit, orbit minimum. All remainders reachable by sign flips; the smallest of them.

Hypercube Q_d. The d-dimensional cube; each orbit is one.

Admissible pattern. A set of shifts H that does not cover all remainders modulo any prime; otherwise one member is always divisible by that prime.

Singular series $\mathfrak{S}(H)$. The product of local correction factors in the Hardy–Littlewood prediction; here the limit of exact vertex counts.

Lifting problem. Showing that admissible configurations are actually realised by primes infinitely often.

Status marks. **[S]** proved, **[G]** exact reformulation, **[N]** numerical, **[H]** heuristic, **[O]** open.

Data and code availability

The reference code is reproduced in Appendix B. All numerical values in this paper were recomputed with an independent Python script (NumPy sieve up to 10^8). This script, its complete

output and the script that generates all figures are provided as supplementary material with this preprint (files `numerics_check.py`, `numerics_output.txt` and `figs.py`; the check runs in under one minute). The earlier papers of the series are openly available under the DOIs listed in the references.

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Conflicts of interest

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